Tree type network revisited

ELL409 | Course assignment report | Prof. Jayadeva

Private GitHub repository
https://github.com/aydecodes/409-treenetwork

Abstract. This report revisits a differentiable tree-type neural network in which every internal node and every leaf is a small neural network. A shared CNN produces image features, routing MLPs compute soft branch probabilities, and leaf MLPs produce class distributions. The approach was evaluated on Cats vs Dogs and CIFAR-10. Near-perfect training accuracy was obtained intentionally on both datasets. Cats vs Dogs retained strong validation accuracy, whereas the CIFAR-10 memorization run made overfitting explicit. The capacity probes did not demonstrate double descent.

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1 Objective and recorded outcomes

The assignment requires a neural tree on Cats vs Dogs and CIFAR-10, a feature strategy, intentional overfitting, a double-descent investigation, node receptive-field visualization, and an attempt to reduce model size. The same soft-tree principle is used for both datasets, with dataset-specific CNN feature extractors and capacity settings.

Dataset and experiment	Training result	Validation or test result	Interpretation
Cats vs Dogs selected run	100% by epoch 68	Best validation 84.63% at epoch 62	High accuracy with later overfitting
Cats vs Dogs size sweep	100% for all five sizes	Best validation 84.26%	32/8 is close to selected accuracy
CIFAR-10 baseline	63.3% at epoch 25	62.0% test	Compact model generalizes reasonably
CIFAR-10 intentional overfit	99.6% at epoch 31	41.2% test; validation near 40%	Severe overfitting
CIFAR-10 capacity probe	22.4% to 30.4%	20.9% to 28.9% validation	No interpolation threshold

The central result is the separation between fitting and generalization. The Cats vs Dogs experiment reaches complete training-set interpolation while retaining strong validation performance. The dedicated CIFAR-10 overfit configuration reaches 99.6% training accuracy but only 41.2% test accuracy.

2 Architecture and implementation

Both models are full binary soft neural trees. A shared CNN maps image x to feature vector f. At internal node j, a routing MLP outputs $r_j(f)$ in (0, 1), the probability of taking the right branch. Each leaf receives the same feature vector and uses its own MLP plus softmax to produce a class distribution. The final prediction is the weighted sum of all leaf distributions, where each leaf weight is the product of left probabilities $1 - r_j(f)$ and right probabilities $r_j(f)$ along its path.

Soft routing is differentiable end to end: every route and every leaf is evaluated during training. A depth-d tree contains $2^d - 1$ routing MLPs and 2^d leaf MLPs, so depth grows node count exponentially.

Feature construction and training procedure

- Cats vs Dogs uses a shared three-block CNN. The selected run used 96 by 96 RGB images, feature size 64, hidden size 16, and depth 1.
- CIFAR-10 baseline uses a compact shared CNN on normalized 32 by 32 RGB images, feature size 64, hidden size 16, and depth 2.
- The CIFAR-10 memorization run flattens the CNN spatial output before projection, preserving more spatial information and raising capacity to 641,323 parameters.
- Adam and negative log-likelihood were used, with fixed seeds, fixed validation splits, validation-selected checkpoints, and progress printing.

3 Cats vs Dogs results

The Cats vs Dogs tree has one routing MLP, two leaf MLPs, and a shared CNN trunk. The selected compact run reaches complete training-set interpolation, while validation peaks earlier.

Feature / hidden / input	Parameters	First 100% train	Best validation	Best epoch
128 / 32 / 128px	not recorded	77	82.95%	90
64 / 16 / 96px	390,469	68	84.63%	62

For the selected 64/16 run, validation accuracy peaks at 84.63% in epoch 62 and training reaches 100% at epoch 68. Validation loss rises later even as training loss continues down, which is the expected signature of overfitting.

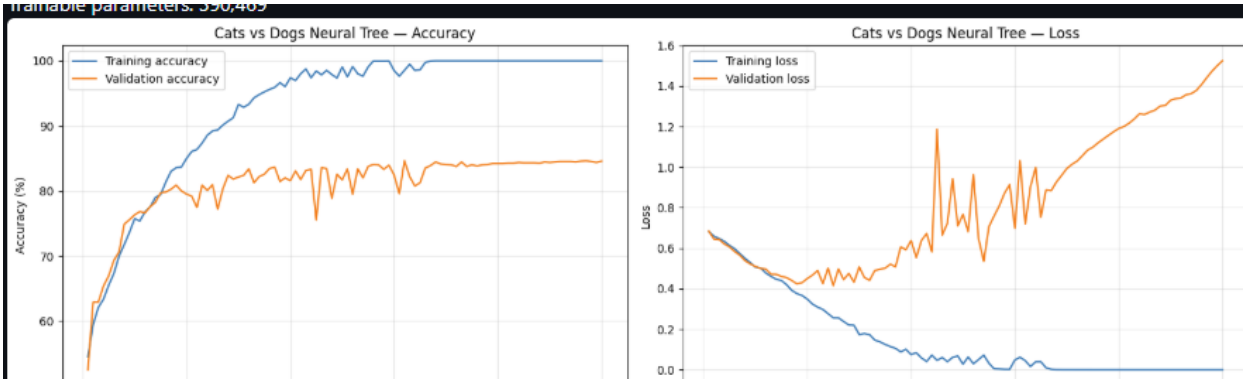


Figure 1. Cats vs Dogs selected-run learning curves. Training reaches 100% while validation accuracy levels near 84% and validation loss rises later.

Capacity and size minimization

Feature / hidden	Parameters	Train accuracy	Validation accuracy	Validation error
16 / 4	375,157	100.00%	83.20%	16.80%
32 / 8	379,877	100.00%	84.26%	15.74%
64 / 16	390,469	100.00%	83.57%	16.43%
128 / 32	416,261	100.00%	83.39%	16.61%
256 / 64	486,277	100.00%	83.39%	16.61%

All five configurations reached 100% training accuracy. The 32/8 model had the lowest recorded validation error and used 2.7% fewer parameters than the selected 64/16 run. The curve does not show double descent: validation error decreases once and then rises slightly, without a later second decrease at larger capacity.

4 CIFAR-10 baseline and intentional overfitting

CIFAR-10 has ten classes and is substantially harder than Cats vs Dogs. The baseline profile uses 20,000 training images, 5,000 validation images, 25 epochs, batch size 128, feature size 64, hidden size 16, and depth 2. It has 68,491 trainable parameters.

Training images	Validation images	Epochs	Final train accuracy	Test accuracy
20,000	5,000	25	63.3%	62.0%

Training and validation curves remain close. The 62.0% test accuracy shows that a compact CNN plus soft tree can learn useful CIFAR-10 structure without severe overfitting in this regime.

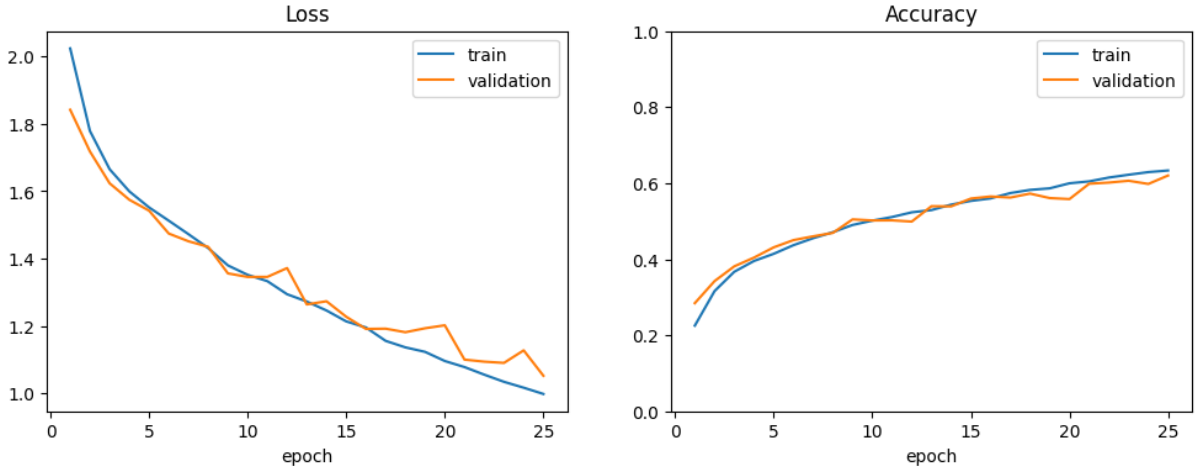


Figure 2. CIFAR-10 baseline curves. Training and validation accuracy remain close over 25 epochs.

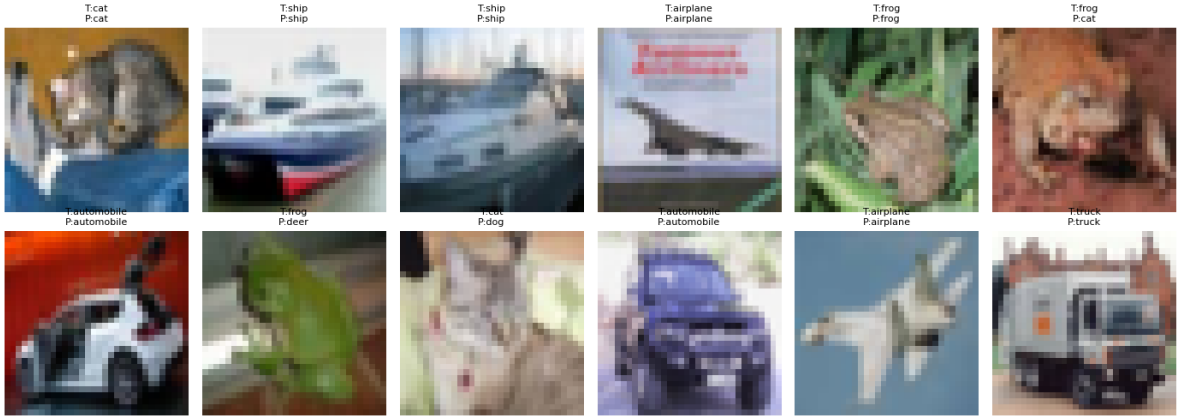


Figure 3. Example baseline predictions. The compact model makes both correct and difficult ten-class errors.

Intentional memorization run

A larger depth-2 model was trained on a fixed 1,000-image training subset and a separate 1,000-image validation subset. Its spatially preserving feature extractor, feature size 128, and hidden size 64 produced 641,323 parameters. It reached 99.6% training accuracy in epoch 31 and stopped at the 99.5% target. Validation remained near 40%, and the best-validation checkpoint obtained 41.2% test accuracy.

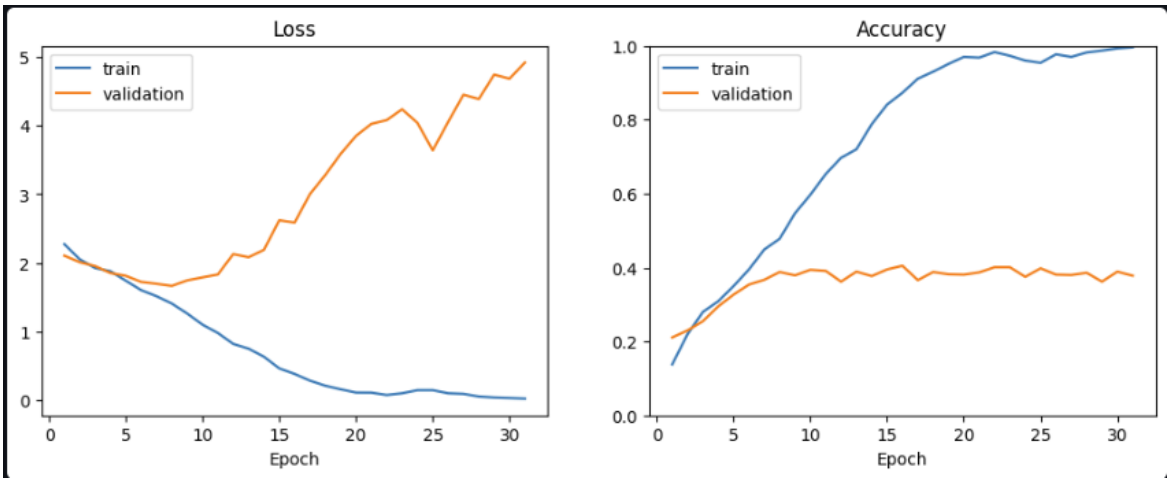


Figure 4. Intentional CIFAR-10 overfitting. Training loss approaches zero while validation loss rises and validation accuracy stays near 40%.

5 Capacity, double descent, and minimization

CIFAR-10 CPU capacity probe

Three configurations were trained for eight epochs on the same fixed 3,000-image training subset and 1,000-image validation subset. This is a controlled CPU-friendly capacity probe, not a definitive double-descent study.

Feature / hidden / depth	Parameters	Train accuracy	Validation accuracy	Validation error
16 / 4 / 1	57,669	22.4%	20.9%	79.1%
32 / 8 / 2	60,635	27.1%	26.9%	73.1%
64 / 16 / 2	68,491	30.4%	28.9%	71.1%

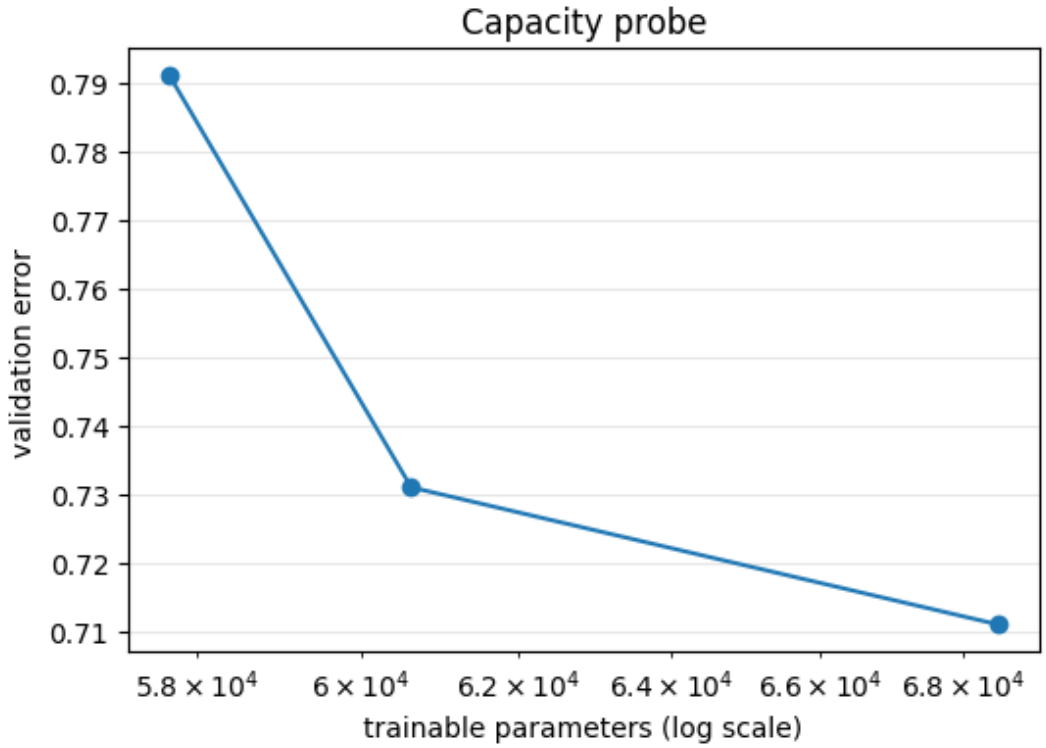


Figure 5. CIFAR-10 validation error against parameter count for the short CPU capacity probe.

Validation error decreases monotonically over the sampled capacities, but no configuration reaches 99% training accuracy. The interpolation threshold is not observed, so the data do not support a double-descent claim. The correct conclusion is that larger capacity improved this limited probe.

Network-size trade-offs

For CIFAR-10, the 60,635-parameter intermediate model removes 11.5% of parameters relative to the 68,491-parameter largest probe, while losing 2.0 validation points. It is a reasonable compact choice only when that loss is acceptable. Otherwise the largest probe is preferred.

What a stronger double-descent study requires

- Six to eight capacities spanning under-parameterized through strongly over-parameterized models.
- Enough training per capacity to reach comparable convergence, while reporting both train and validation error.
- A fixed small training subset and validation split so that interpolation is computationally reachable.
- Multiple random seeds before claiming a non-monotonic effect.

6 Node receptive fields and routing interpretation

A CNN has spatial receptive fields because convolutional filters operate on local image neighborhoods. The routing and leaf MLPs do not have spatial receptive fields themselves: they consume the CNN feature vector. Node interpretation therefore uses gradient saliency of a routing score with respect to the input image. Brighter pixels mark locations where a small perturbation has a larger local effect on routing evidence.

For Cats vs Dogs, saliency is computed from the routing logit rather than its sigmoid probability. This avoids vanishing gradients when a confident router outputs a probability very close to zero or one. The maps are sensitivity explanations, not causal object masks.

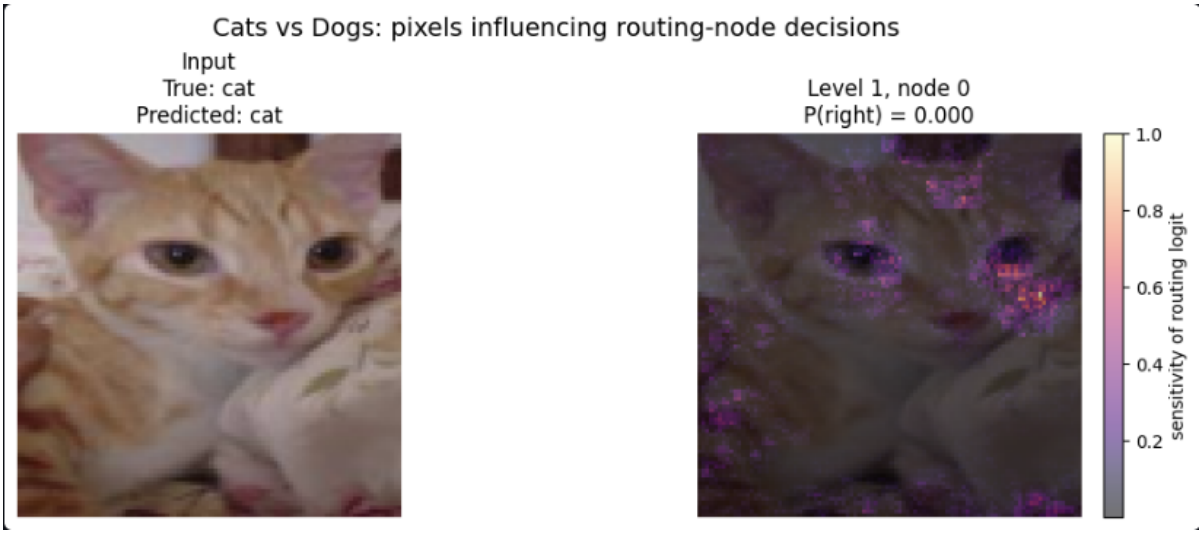


Figure 6. Cats vs Dogs routing-node saliency. Bright pixels influence the root node's routing evidence.

For CIFAR-10, all routing MLPs are evaluated because routing is soft. Leaf reach probabilities weight each leaf's class distribution in the final prediction. The map below is an effective routing-sensitivity visualization, not a literal MLP receptive field.

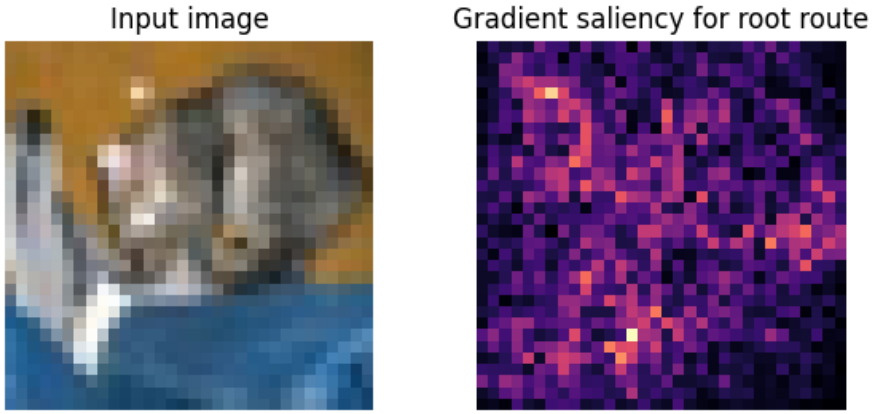


Figure 7. CIFAR-10 input image and root-routing gradient saliency.

CNN filters and observations

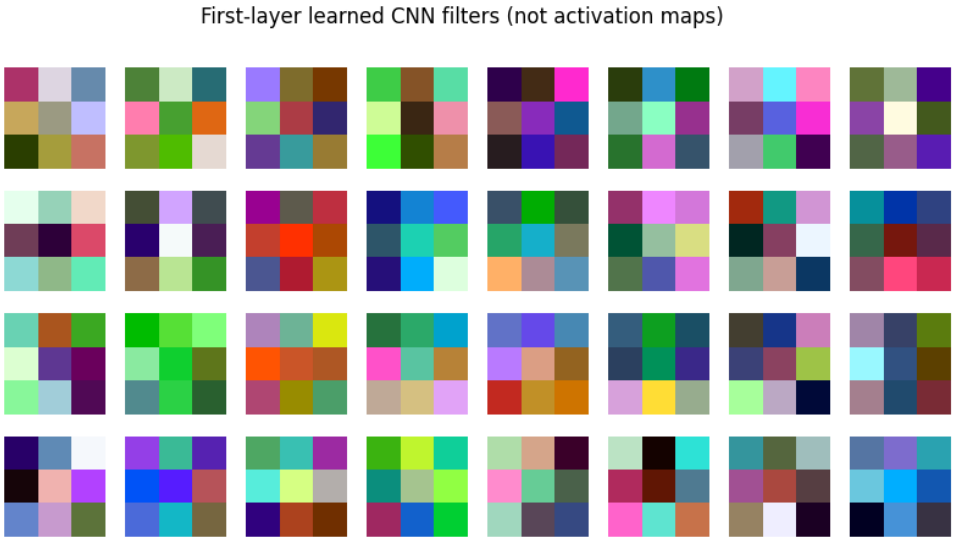


Figure 8. First-layer learned CNN filters. These are learned filters, distinct from activation maps and saliency maps.

Observed interpretation

- Soft routing gives all leaves a role, but high-probability paths contribute more to the final prediction.
- A saturated Cats vs Dogs router can still be inspected through gradients of its pre-sigmoid logit.
- The CIFAR-10 maps are diffuse, consistent with compact features and low-resolution ten-class recognition; they should not be read as precise object segmentation.

7 Conclusions and future directions

Conclusions

A tree in which every internal node and every leaf is a neural network can be trained end to end on image data using soft routing. A shared CNN is an effective feature mechanism because every node can use the same learned representation. Cats vs Dogs reaches 100% training accuracy while retaining strong validation performance. The CIFAR-10 dedicated memorization run reaches 99.6% training accuracy with much lower validation and test accuracy, directly demonstrating overfitting. The normal CIFAR-10 baseline instead reaches 62.0% test accuracy with a small train-validation gap.

Neither experiment showed double descent. In Cats vs Dogs, making the model larger did not lead to another improvement after the first improvement. In the short CIFAR-10 sweep, none of the models learned the training data fully. Therefore, double descent was investigated but was not seen in these runs.

Future directions

- Test more small, medium, and large tree models to check whether double descent appears.
- Compare a shallow tree with a deeper tree to see whether extra decision nodes help.
- Try simple regularization methods, such as image augmentation or early stopping, and compare the accuracy.
- Visualize more images to see what parts of an image each routing node focuses on.